index | Status: current | App v1.0.20 | Reviewed 2026-10-24

Google AI Edge Gallery — Engineering Documentation

Google AI Edge Gallery is an Android app (label "Edge Gallery") that runs open-weight generative AI models **fully on-device** with LiteRT-LM, Android AICore, and MediaPipe. This folder describes how it is built. For what the product does from a user's point of view, see the root README.

Documents

#	Document	id	What it answers
01	Architecture	`architecture`	How the app is layered, how modules connect, startup and navigation
02	Agents	`agents`	How the in-app agent runtime, tools, Agent Skills, MCP, and intents work
03	Functional design	`functional-design`	What each feature does, its user flow, inputs, limits, and models
04	Technical design	`technical-design`	Model catalog, download, runtimes, sessions, persistence, workers, telemetry
05	Tech stack	`tech-stack`	Languages, build tooling, libraries, and versions
06	Testing	`testing`	Current test state (none automated) and the proposed strategy
07	Code conventions	`code-conventions`	Conventions observed in the code
08	How-to guides	`how-to-guides`	Step-by-step recipes for common engineering tasks
09	Requirements	`requirements`	Goals, non-goals, functional and non-functional requirements
10	Security & privacy	`security-privacy`	Permissions, data at rest, network surface, risks
11	Build, CI & release	`build-ci-release`	Build variants, CI workflows, versioning, release steps
12	Architecture decisions	`architecture-decisions`	ADRs reconstructed from the code
13	Glossary	`glossary`	Domain and project terms
14	Known issues	`known-issues`	Defects, risks, and doc/code mismatches found during review
—	AGENTS.md	`agents-md`	Instructions for AI coding assistants working in this repo

Reading order for a new engineer: 01 → 05 → 03 → 04 → 02 → 07 → 08. For a reviewer or auditor: 09 → 10 → 14.

Existing docs (not moved)

File	Topic
README.md	Product overview and install links
DEVELOPMENT.md	Hugging Face OAuth setup for local builds
Function_Calling_Guide.md	Adding a Mobile Actions function
Bug_Reporting_Guide.md	Capturing Android bug reports
skills/README.md	Authoring Agent Skills (`SKILL.md` , JS skills)
mcp/README.md	Connecting MCP servers

Conventions used in these docs

Front matter schema

Every document starts with YAML front matter so tools can index it:

Key	Type	Meaning
`id`	string	Stable identifier. Other docs refer to it in `related` .
`title`	string	Human title.
`doc_type`	enum	`index` , `architecture` , `agents` , `functional-design` , `technical-design` , `tech-stack` , `testing` , `code-conventions` , `how-to` , `requirements` , `security` , `release` , `adr` , `glossary` , `known-issues` , `agent-instructions`
`status`	enum	`current` = describes code as it is. `proposed` = recommendation, not implemented. `mixed` = both; each section is labelled.
`app_version`	string	`versionName` the doc was checked against (from `Android/src/app/build.gradle.kts`).
`last_reviewed`	date	ISO date of the last check against the code.
`audience`	list	Intended readers.
`related`	list	`id` s of related docs.
`sources`	list	Repo-relative paths the doc was derived from. Re-check these when updating the doc.

Path shorthand

Kotlin sources live under `Android/src/app/src/main/java/com/google/ai/edge/gallery/`. In tables and prose this prefix is written as **`gallery/`**, e.g. `gallery/agent/AgentRuntimeExecutor.kt`. Front matter `sources` always use full repo-relative paths.

Labels in text

- **[current]** — verified in code at `app_version`.
- **[proposed]** — a recommendation; nothing in the repo implements it yet.
- **[inferred]** — deduced from indirect evidence (comments, naming, build files); not directly confirmed.
- **[to verify]** — could not be confirmed from the source in this repo.

Keeping docs current

When a change touches a file listed in a doc's `sources`, update that doc and its `last_reviewed` date in the same change. If you bump `versionName`, re-check `app_version` across docs.